

Task 3 Report: Deep-Dive Analysis & Interactive Dashboard

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Tool: Microsoft Power BI

1. Introduction

This project focused on performing a deep-dive analysis of a retail sales dataset and developing an interactive Power BI dashboard. The dashboard enables users to monitor business performance using KPIs, charts and slicers.

2. Objectives

- Define key performance indicators (KPIs).
- Perform business-focused analysis.
- Create an interactive dashboard.
- Support data-driven decision making.

3. Dataset Overview

The dashboard uses the cleaned retail sales dataset prepared in Task 2.

Attribute	Type	Purpose
Order_ID	Categorical	Order tracking
Order_Date	Date	Trend analysis
Customer_ID	Categorical	Customer analysis
Customer_Name	Text	Customer identification
Age	Numeric	Demographics
Gender	Categorical	Segmentation
City	Categorical	Regional analysis
Product	Categorical	Product analysis
Category	Categorical	Category analysis
Quantity	Numeric	Units sold
Unit_Price	Numeric	Pricing
Total_Sales	Numeric	Revenue

4. Dashboard Development Process

- Imported the dataset into Power BI.
- Validated and transformed data using Power Query.
- Created DAX measures for KPIs.

- Designed KPI cards, charts and slicers.
- Applied a consistent dashboard layout and theme.

5. Key Performance Indicators

KPI	Calculation	Purpose
Total Sales	SUM(Total_Sales)	Overall revenue
Total Quantity	SUM(Quantity)	Units sold
Total Unit Price	SUM(Unit_Price)	Pricing metric
Total Orders	COUNT(Order_ID)	Order volume
Total Customers	DISTINCTCOUNT(Customer_ID)	Customer reach

6. Dashboard Visualisations

- KPI Cards
- Monthly Sales Trend
- Sales by Product
- Orders by Category
- Revenue by Category
- Gender Distribution
- Quantity by Product
- Age and City Slicers

7. Business Insights

- Monthly trends reveal peak and low sales periods.
- Top-performing products and categories are easily identified.
- Customer demographics support targeted marketing.
- Interactive filters improve analytical flexibility.
- KPIs provide a quick overview of business performance.

8. Tools & Technologies

- Microsoft Power BI
- Power Query
- DAX
- Microsoft Excel
- Business Intelligence

9. Conclusion

The dashboard successfully converts transactional data into meaningful business insights through interactive visualisations, KPIs and filtering. It demonstrates practical skills in Power BI, DAX, dashboard design and business intelligence.

10. Future Scope

- Publish dashboard to Power BI Service.
- Add forecasting and predictive analytics.
- Create drill-through reports.
- Integrate live database connections.